VALID — a 12-item validation checklist for financial machine learning

orcajae.github.io/valid-framework

Twelve binary items in a fixed order: the statistical gate first, the economic gate second.

Stage 1 · Statistical

verifiable from the manuscript alone

- [ ] **V1. Report prediction class distribution**
  Do you report the class distribution of your model's predictions?
- [ ] **V2. Test with/without class balancing**
  Do you test both with and without class balancing?
- [ ] **V3. Use temporal splitting only**
  Do you split strictly by time, with purge and embargo gaps?
- [ ] **V4. Apply CPCV with PBO**
  Do you apply CPCV and report the probability of backtest overfitting?
- [ ] **V5. Report Var(SR_IS)**
  Do you report the variance of in-sample Sharpe across configurations?
- [ ] **V6. Include permutation tests (≥100)**
  Do you run a permutation test with at least 100 iterations?

Stage 2 · Economic

required before anything is traded

- [ ] **V7. Report net performance with costs**
  Do you report performance net of transaction costs?
- [ ] **V8. Cost sensitivity analysis**
  Do you show how conclusions hold across a range of cost assumptions?
- [ ] **V9. Compare against simple baselines**
  Do you compare against simple active baselines, not only buy-and-hold?
- [ ] **V10. Evaluate in bear markets and regime transitions**
  Do you evaluate in bear markets and regime transitions?
- [ ] **V11. Report trade frequency**
  Do you report trade frequency?
- [ ] **V12. Provide code for reproducibility**
  Is your code available so the result can be reproduced?

How to score

One point per item, satisfied or not — no partial credit. Score Stage 1 and Stage 2 separately, then total out of 12.

What the total means

Each unchecked item is a way a backtest can look better than the strategy is. A study that clears Stage 1 but fails Stage 2 has a statistical signal with no exploitable alpha.

What it is not

Self-scoring is not a certification. It records your own reading of your own artifacts; check each item against the code and data themselves. Most empirical results of the accompanying paper were withdrawn in October 2026 (CHANGELOG E13).

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